Xuanzhi Chen

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Website | GitHub | Google Scholar | LinkedIn

Education

Peking University Beijing, China
B.Eng. Candidate in Bioinformatics, School of Life Sciences *Sep. 2023 – Jun. 2027 (Expected)*

- **GPA: 3.77/4.0.** Pursuing a double degree in Economics at the National School of Development.
- **Selected coursework:** Computational Immuno-oncology (99/100); Data Structures and Algorithms (98/100); Statistical Analysis of Genomics Data (A); AI Methods and Applications in Life Sciences (A).

Preprint

Nianping Liu, **Xuanzhi Chen**, Miao Cui, Xiaoying Liao, Xiaoke Song, Sairam Pantham, Weize Xu, and Xiaojie Qiu. **Bio-Babel: autonomous cross-language reconstruction of agent-ready computational biology ecosystems.** *bioRxiv*, 2026. https://doi.org/10.64898/2026.08.30.743167.

Research Experience

College of Future Technology, Peking University Beijing, China
Undergraduate Researcher; Advisor: Prof. Hu Zeng *Jul. 2024 – Present*

STATES-GBM: Spatial transcriptomics and translatomics in glioblastoma

- Benchmarked cell segmentation, integrated 3D Cellpose nuclear segmentation into ClusterMap, and optimized image stitching for STATES. Reduced seams misaligned by > 3 pixels from 14.2% to 0.3% and multi-nucleus segments from 28.8% to 13–14% against the same nuclear reference.
- Evaluated and refined segmentation using image inspection and downstream cell annotation. Analyzed malignant-cell identities, subtypes, and spatial variation in translation efficiency in glioblastoma.

iST Agent: Automated execution of imaging-based spatial transcriptomics workflows

- Built an agent for imaging-based spatial transcriptomics that separates source-derived workflow contracts from memory of execution failures and fixes.
- Evaluated the agent on MERlin using public MERFISH data. Contracts more than halved failed cluster jobs for both models tested; reviewed memory further reduced the smaller model's token-based cost by 16% compared with contracts alone, with no significant difference in task success.

Department of Genetics, Stanford University Stanford, CA
Undergraduate Researcher; Advisor: Prof. Xiaojie Qiu *Jun. 2026 – Present*

On-site summer research, Jun.–Aug. 2026; continued remotely.

Bio-Babel: Agent-ready computational biology software

- Developed an annotation agent that generates machine-readable usage specifications for reconstructed packages. Extended coding-agent compatibility and built tooling for agent-driven Python-to-C++ reconstruction.
- Conducted the pancreatic differentiation case study, including preprocessing, cell annotation, and analyses using Bio-Babel's agents. Reviewed trajectory analyses and performed genotype-aware screening to interpret reduced maturation signatures under SWI/SNF loss.
  UMI-tools-cpp | Pancreatic differentiation case study

iST upstream model

- Developing and training a model on public MERFISH data to jointly localize RNA molecules, decode barcodes, segment cells, and assign molecules to cells.
- Combining RNA fluorescence and cellular morphology to learn cross-round correspondences without separate image warping; testing whether joint modeling reduces errors propagated by sequential processing.

Canchen Technology Remote
Research Intern; Mentor: Dr. Lei Li *Oct. 2026 – Present*

Agentic world models for virtual tissues

- Developing a research proposal combining foundation models, mechanistic models, and LLM agents for tissue perturbation prediction. Proposed feedback between biological reasoning and quantitative model computation, and outlined comparisons to evaluate model integration, agent contributions, and iterative feedback.

Selected Presentations

Building an Agentic Tissue World Model.
Journal club presentation, Canchen Technology, Nov. 2026.

Bio-Babel: autonomous cross-language reconstruction of agent-ready computational biology ecosystems.
Invited oral presentation, PKU Undergraduate Honors Program in Biology Annual Symposium, Oct. 2026.

Bio-Babel and iST Upstream Model.
Research presentation, Qiu Lab, Stanford University, Aug. 21, 2026.

How to Build the Virtual Cell?
Group course presentation, Systems Biology, Peking University, Jun. 6, 2026.

SpatialFusion: A Lightweight Multimodal Foundation Model for Pathway-Informed Spatial Niche Mapping.
Journal club presentation, Zeng Lab, Peking University, May 27, 2026.

Honors and Awards

Academic Excellence Award, Peking University	2026
Peking University Scholarship	2025
Peking University Freshmen Scholarship	2023
Gold Medal, China National Biology Olympiad Final	2022

Leadership & Activities

Volleyball Enthusiasts Association, Peking University

Chair	2024–2025
Director of Competitions	2023–2024

- Organized Peking University's autumn Freshman Cup and spring PKU Cup volleyball tournaments.
- Coordinated winter and summer training camps and handled association finances.

School of Life Sciences Volleyball Team, Peking University

Captain	2025–2026

Technical Skills

Programming & machine learning: Python, R, PyTorch.

Bioinformatics & imaging: Scanpy, Seurat, Cellpose, ClusterMap.

Computing & automation: Linux, Slurm, Git; scientific agents and workflow automation.

Experimental: In situ spatial omics (STATES), molecular cloning, CRISPR-Cas9, PCR/qPCR, Western blotting.